

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Technology (CSE)

QUIZ #3 (SET-B)

DURATION: 20 MINUTES

SUMMER SEMESTER, 2023-2024

FULL MARKS: 15

CSE 4711: Artificial Intelligence

Programmable calculators are not allowed.

Answer **all 2 (two)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

Student ID:

1. Consider a grid world Markov Decision Process (MDP) shown in Figure 1 with deterministic actions (all actions chosen will be executed, unless the agent is blocked by a wall). The terminal states are marked with their reward values +10 and +1. The action space $\mathcal{A}:\{\text{left, right, up, down, stay}\}$. The discounting factor $\gamma = 1$, and there is no living reward. Applying value iteration, one can find the values of the states.

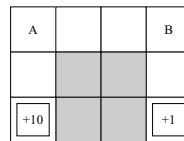


Figure 1: A simple grid world for Question 1.

- a) Find the smallest values of k for which $V_k^*(B) > 0$. What is the smallest value of k for which $V_k^*(B) = V^*(B)$? Also show the values of $V_k^*(B)$ and $V^*(B)$. 3 + 3
(CO1)
(PO1)

Solution:

- For $k = 3$, $V_k^*(B) = 1$.
- For $k = 6$, $V_k^*(B) = V^*(B) = 10$.

Rubric:

- 2 point for correctly finding each k .
- 1 points for correctly finding each values.

- b) Assume all actions succeed 80% of the time or fail otherwise (remains in the same state). What is the smallest value of k for which $V_k^*(A) = V^*(A)$? What is $V^*(A)$? 3
(CO1)
(PO1)

Solution:

- The lowest value of k is ∞ . Because the transition is non-deterministic, it's not guaranteed this reward can be collected in 3 steps. It could any number of steps from 3 through ∞ .
- $V^*(A) = 10$.

Rubric:

- 2 point for correctly finding each k .
- 1 points for correctly finding each values.

2. Assume that you have \$100 and you are passing through Mohammadpur. There is an 20% chance that your money gets stolen. Your utility function is $U(x) = \log_e(1 + x)$, where x is the amount of money you have left.

a) What is your expected utility after going through Mohammadpur?

2
(CO1)
(PO1)

Solution:

$$0.8 \times \log_e(1 + 100) + 0.2 \times \log_e(1 + 0) = 3.6921$$

Rubric:

- 2 points for correctly calculating the value.

b) If you pay an premium of \$30 to an insurance company, they will reimburse you \$70 if your money gets stolen. Should you take this offer from the insurance company?

4
(CO1)
(PO1)

Solution:

Whether your money gets stolen or not, you will always have to pay \$30 to the insurance company. Your new expected utility would be: $0.8 \times \log_e(1 + 70) + 0.2 \times \log_e(1 + 70) = 4.2627$, which is greater than the expected utility of not taking the insurance. Thus, you should take the insurance offer.

Rubric:

- 3 points for correctly calculating the value.
- 1 point for correctly choosing.